

Role Title:	Specialist Materials Engineer - Experimental	Division / Function	Fusion Technology Division
Grade:	F	Reports to: (job title)	Applied Materials Technology team leader/Group lead
Skill Type:	Engineering	Line Manager (yes / no)	No

Overall Purpose:

- Why does the role exist?
- What does the role aim to achieve?
- What wouldn't happen if the role didn't exist?
- What does the role contribute to or deliver for UKAEA? (consider UKAEA mission and goals, and Divisional goals and high-level objectives).

The vision of the Applied Materials Technology (AMT) Group is to become the global leader in the provision of validated material property data – and its associated R&D technologies – consistent with the requirements of current and future fusion plant. This vision is convergent with that of the Fusion Technology (FT) Division, which is to enable the development and qualification of fusion device equipment for service in commercial fusion power plant.

The Specialist Materials Engineer – Experimental is a recognised subject matter specialist who works mainly independently, beginning to review and approve technical outputs within the area of expertise. They contribute to the realisation of AMT vision through delivery of independent moderate/complex research work, applying best practice in the application of materials testing technologies – particularly those relying on measurement techniques for materials performance assessments. They are qualified practitioners with a sound understanding of core subject matter pertaining to materials engineering. They act as a specialist advisor in own area of expertise and strive to develop novel research methodologies pertaining to materials engineering.

• **Accountabilities:**

- Capture important tasks that need to be carried out to ensure that the role is successful in achieving its purpose. What is the deliverable/output/end result?
- Use action verbs e.g. act as, create, develop, drive, forecast, implement, analyse, compile, escalate.
- Include the level at which the role is supporting, coordinating, advising, managing, leading, etc., and the output of these actions.
- Consider the scope, level, and complexity of the problems or deliverables the role will address, and what freedom does the job holder have to solve them (e.g. are they following procedures? Applying/ interpreting legalisation? Do they advise/decide on options available, is there a high level of ambiguity - do they start from first principles?)
- Consider the breadth and depth of impact the role has – individual contributor/team/unit/ group/ division/organisation?
- Consider how much autonomy the role has? what latitude is permitted within the role to make decisions and determine the method of delivery to achieve the desired end results – where does the buck stop, who makes the final decisions, what is the budget and the autonomy of how the budget is spent?

1. Identify, plan, design, develop and perform medium/high-complexity work independently, to support delivery of projects where application of materials testing technologies – particularly those related to diagnostics, component-level testing, similar and dissimilar materials joints, and miniaturised testing techniques – are critical to mission success.
2. Act as a specialist advisor in the field of materials testing technologies, to provide advice to projects requiring such guidance.
3. Own specific experimental elements within projects, independently deciding route for preparation, experimental parameters and execution within agreed work package requirements, managing time and associated budgets, escalating issues to senior colleagues.
4. Develop proficiency in scientific computing (e.g Python, Matlab, data processing, optimisation, etc.) to ensure work is performed efficiently and in line with AMT good practice.
5. Lead and contribute to the dissemination of research outputs (via technical reports, publications, invited talks, etc.) that focus on application of materials testing technologies, to maximise research impact, establishing an active external technical profile through conference participation, publication and engagement with relevant professional networks.
6. Work towards becoming an expert in the field(s) of Materials Testing 2.0, digital-physical interfaces, synergistic testing and/or testing in fusion-relevant environments, to enable scalable measurement of material performance in the physical space.
7. Apply and actively promote FAIR data principles across own work and within the immediate team, taking responsibility for raising data management standards and ensuring outputs are reusable across UKAEA.
8. Build and maintain working relationships with external collaborators and academic partners, contributing independently to collaborative technical activities and developing an external technical profile that raises the group's visibility.
9. Provide structured technical mentoring, upskilling and knowledge transfer to staff, students, secondees and other team members, taking a deliberate approach to developing their competence across both computational and experimental aspects of the work.
10. Familiarise yourself with activities across UKAEA and with external stakeholders, including academic collaborators and industrial end users, to establish a technical support network that will support your delivery activities.
11. Promote UKAEA's values, championing a culture of equality, diversity, and inclusion.
12. Ensure that your behaviour promotes the safety, health and welfare of yourself and your colleagues.
13. Contribute to improving best practices and quality standards within the team, division and across the organisation.
14. Budget Responsibility: You may be required to manage budgets up to £50k allocated to your specific tasks, working alongside the budget holder.

Specific Qualifications/Experience:

- Capture the minimum required relevant qualification needed to deliver the outputs of the role.
- Refer to the extent of experience, job knowledge, technical skills, specialist knowledge e.g. demonstratable, track record of, extensive, familiar with, proficient, advanced, expert, etc.
- Include the soft skills required to successfully perform in the role e.g. leadership, communications, adaptability, build rapport, planning, prioritising, problem-solving, creative thinking, etc.

Essential:

- PhD in a STEM-related discipline or substantial relevant industrial experience
- Practical experience with one or more of: mechanical testing; physical testing; miniaturised testing; classical strength of materials; mechanisms leading to material failure (e.g., creep, fatigue, fracture).
- Demonstrated Experience of following FAIR data principles (e.g., adoption of Git and Zenodo) in performing your work.
- Demonstrated experience applying FAIR data principles in a research setting.
- Experience working in a collaborative, cross-functional team environment, contributing to the development of less experienced colleagues
- Excellent verbal and written communication skills, particularly with regards to writing concise and impactful technical reports and scientific papers.
- Ability to use initiative and creativity to resolve queries or issues and make improvements in a proactive manner and apply appropriate attention to detail.
- Good planning, organising, and time management skills including the ability to prioritise workload.
- Demonstrably proficient in the Microsoft Office Suite, in particular Word, Excel and PowerPoint.

Desirable:

- Proven experience of working in a fusion related field.
- Experience of scientific programming in scripting languages (e.g., Python, Matlab) and compiled languages (e.g., C++, Fortran).
- Familiar with materials modelling and simulation, their benefits/limitations, and the uncertainties associated with the data produced via their associated technologies.
- Working towards chartered status with a relevant professional body, with active and evidenced progress.

Additional Duties:

- Include specific appointments e.g. Responsible Officer, Authority to Operate Holder,
 - Include if the role needs to be on-call/do shift work, wear PPE, deputise for, support on.
 - Consider if the role needs serve on advisory boards/committees or to represent at forums, conferences.
 - Consider if the role has to undertake national or international travel.
 - Include if qualifications need to be undertaken and completed specifically required for the role.
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- Must complete all required safety qualifications and role-specific training.
 - May be asked to oversee and provide structured guidance to secondees, postdoctoral researchers or students on specific assigned activities.
 - May be required to contribute to or lead elements of conference presentations and external engagement activities.
 - Contribute to the identification of additional collaboration opportunities

